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AI-Based Automated Grading of Short Answer Responses with Interactive Feedback

Ali Heydari[a], Saeed Alkendi[a], Dr. Sauleh Eetemadi[a,*]

[a]School of Computer Science, University of Birmingham, Birmingham B15 2TT, United Kingdom

Abstract

In this paper, we discuss our work with LLM models for an automated grading system for short answer responses along with generating interactive feedback. We used a locally fine-tuned GPT-2 model and an API-based pipeline using GPT-4o vision plus LLaMA-2 models for retrieval augmented generation. The system processes scanned exam papers, extracts student responses, assigns scores, and generates feedback. We used the students' data with their explicit consent for this research and processed their anonymized university exam papers from the Current Topics In Data Science course with the fine-tuned GPT-2 model and we achieved high predictive accuracy (R^2 = 0.895, MAE = 0.42). However, we found the API-based pipeline more effective in terms of handling complex responses and feedback quality. We also highlighted the trade-offs between computational efficiency and feedback quality in this system.

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Keywords: Automated grading; Large language models; Transformer models; Educational assessment; Feedback generation; Retrieval-augmented generation

1. Introduction

Educational assessment faces significant challenges in manual grading of short answer responses, including time consumption, inter-rater inconsistency, and delayed feedback delivery [7]. Traditional automated scoring systems have evolved from rule-based keyword matching to statistical methods and machine learning approaches, but often lack explanatory capabilities and struggle with semantic understanding [4].

Recent advances in transformer architectures and retrieval-augmented generation (RAG) offer new opportunities for developing automated grading systems that provide both accurate scores and meaningful feedback [6]. However, existing systems typically prioritize either efficiency or feedback quality, rarely achieving both simultaneously.

This research addresses these limitations by developing and evaluating two complementary automated grading pipelines: a locally fine-tuned GPT-2 model for efficient scoring and an API-based pipeline using advanced language models with RAG for rich feedback generation. Our contributions include: (1) implementation of dual grading archi-

* Corresponding author. Tel.: +97-158-544-6114
E-mail address: s.eetemadi@bham.ac.uk

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tectures balancing efficiency and quality, (2) integration of OCR and structure detection for scanned exam scripts with RAG framework, and (3) comprehensive evaluation using both statistical metrics and feedback quality assessment on a dataset of 64 student papers.

2. Related Work

2.1. Early Automated Grading Methods

Early automated grading systems employed rule-based approaches and keyword matching techniques. Systems like C-rater [4] used predefined rules and synonym recognition, while statistical methods like Latent Semantic Analysis (LSA) measured semantic similarity between student responses and reference answers [5]. These approaches were limited by their inability to capture nuanced reasoning and contextual understanding.

2.2. Machine Learning Approaches

Feature-based machine learning models included lexical, syntactic, and semantic features using regression or classification algorithms [7]. Neural network methods, especially Recurrent Neural Networks (RNNs) and Convolutional Neural Networks (CNNs), learned text representations directly from data. This improved how well they worked across different fields [2].

2.3. Transformer Architectures and RAG

Transformer-based models like BERT [1] and GPT have changed automated assessment through transfer learning. Fine-tuning pretrained models on educational data has reached top performance [8, 3]. Retrieval augmented generation (RAG) frameworks combine the retrieval of relevant rubric information with generative models to generate scores and explanations [6]. This improves transparency and matches the grade standards [6].

3. Methodology

3.1. System Architecture

As shown in Figure 1 we created two grading pipelines that work well together:

Local Pipeline: A fine-tuned GPT-2 model trained on response-score pairs for direct prediction. This pipeline works independently, protecting privacy and keeping costs low after initial training.

API-based Pipeline: This one uses GPT-4o Vision for OCR and structure extraction, along with LLaMA-2 models for evaluation. It uses cloud-based models through APIs, allowing it to handle various response types effectively and generate detailed feedback with RAG elements.

3.2. Data Preparation

The dataset contained two different exam papers from the Current Topics in Data Science course with their short response. The first exam had 33 students and the second one had 31 students. Total of 64 student's papers containing 256 pages were evaluated.

All student data was anonymized and used with explicit consent obtained through institutional permission forms prior to analysis, ensuring ethical compliance with data handling.

All papers were scanned into PDF format and processed in multiple stages.

- PDF-to-image conversion and optical character recognition (OCR) using GPT-4o Vision
- Identification of the question structure from the model answer sheets
- Alignment with official grade sheets using submission IDs
- Construction of training samples in conversational format with system prompts, student responses, and target scores with feedback

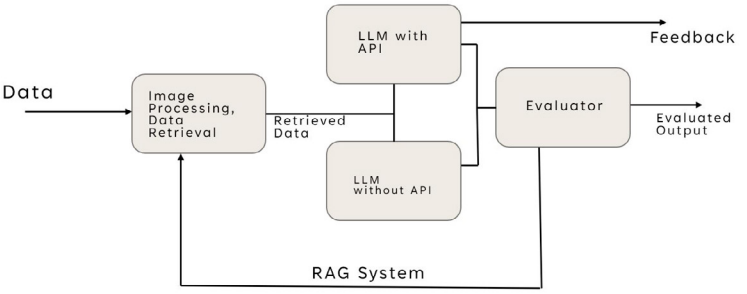


Fig. 1: Methodology overview, complete Pipeline of workflow

3.3. Model Training and Configuration

The GPT-2 model was fine-tuned using a combined loss function:

$$\mathcal{L} = \lambda \mathcal{L}_{\text{LM}} + (1 - \lambda)\mathcal{L}_{\text{score}}$$

where $\mathcal{L}_{\text{LM}}$ is the language modeling loss for feedback generation and $\mathcal{L}_{\text{score}}$ is the mean squared error for scoring prediction. The parameter $\lambda = 0.7$ balanced text generation quality and scoring accuracy.

Note: Specific training hyperparameters (learning rate, batch size, epochs) were not detailed in the source material and would be required for full reproducibility.

The API-based pipeline used structured prompts with RAG framework, retrieving relevant exemplar responses and rubric information to ground the evaluation process. The retrieval component used a dense vector similarity search against a corpus of model answers and rubric components, typically retrieving the top 2-3 most relevant passages for augmentation.

3.4. Prompt Engineering

We developed specialized prompt templates for different model types:

LLaMA Models System Prompt:

```
"You are an expert exam grader. Provide clear, structured responses."
```

LLaMA Models User Prompt Template:

```
You are grading an exam. Be flexible and accept alternative correct approaches.
QUESTIONS TO GRADE:
[Question list with point values]
MODEL ANSWER (showing one possible approach):
[Reference text]
STUDENT ANSWER:
[Student response text]

INSTRUCTIONS:
```

```
1. Grade each question listed above
2. Give credit for correct alternative solutions
3. Use EXACTLY this format for EACH question:
Q1: 8/10
Explanation: [Brief reason for the score]
Q2: 7/15
Explanation: [Brief reason for the score]
Start grading now:
```

GPT-4o Models System Prompt:

```
IMPORTANT GRADING INSTRUCTIONS:
1. Be STRICT but FAIR in grading - this is a formal examination
2. Deduct points for missing steps, computational errors, incomplete answers
3. Use partial credit guidelines
4. DO NOT give full marks unless the answer is COMPLETELY correct
5. Compare CAREFULLY with the model answer for completeness
```

GPT-4o Models User Prompt Template:

```
[Grading rubric with question point values]

Return a JSON object with this EXACT structure:
{
    "questions": [
        {
            "id": "Q1",
            "points": 10,
            "details": [
                {
                    "id": "Q1",
                    "awarded_points": 7,
                    "max_points": 10,
                    "explanation": "Correct approach but missing final step."
                }
            ]
        }
    ]
}
```

Explanation Generation System Prompt:

```
"You are a helpful teaching assistant explaining exam grades. Remember that the model answer shows
ONE possible solution, but students may have used different valid approaches. When explaining grades,
acknowledge when alternative correct methods were used. Be specific about what was done well and
what could be improved."
```

3.5. Evaluation Framework

We evaluated system performance using:

- Accuracy metrics: R^2, MAE, RMSE, Cohen's κ
- Feedback quality: Specificity, groundedness, and helpfulness (Likert scale)
- Computational efficiency: Training time, inference latency, and resource requirements

The dataset was split into training (70%), validation (15%), and test (15%) sets for the offline GPT-2 model. API-based models were evaluated in zero-shot and few-shot settings on the entire dataset.

4. Experiments

4.1. Dataset and Environment

Experiments utilized 64 anonymized exam scripts from a Current Topics in Data Science course, comprising two separate exams (33 papers from Exam 1 and 31 papers from Exam 2). In total, approximately 256 pages of exam content were processed through the grading pipeline. All responses were manually graded by a single experienced instructor to establish ground truth scores.

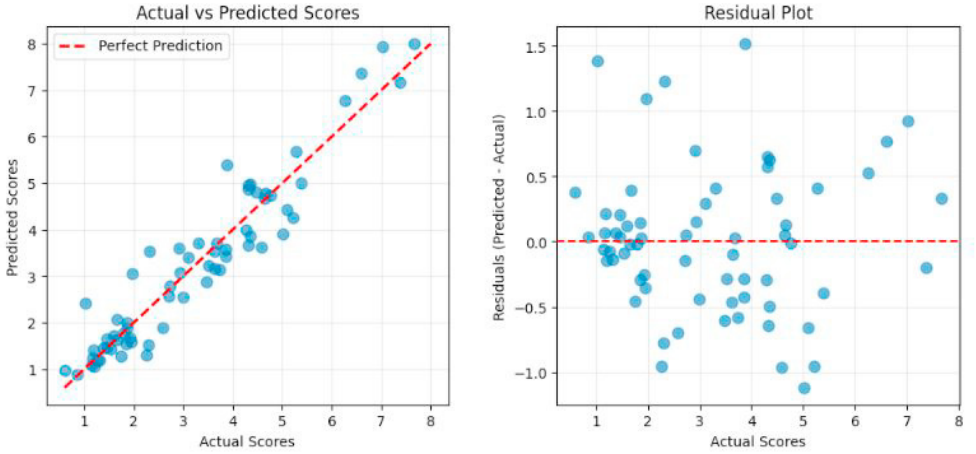


Fig. 2: Differences between predicted and actual scores for the offline GPT-2 fine-tuned model

The offline pipeline ran on a workstation with NVIDIA RTX 3060 GPU, while the API-based pipeline utilized cloud services through a custom Streamlit interface.

4.2. Experimental Configurations

We evaluated three system configurations:

- **Offline GPT-2:** Locally fine-tuned model
- **GPT-4o + LLaMA-2:** API-based pipeline with GPT-4o Vision and LLaMA-2
- **Groq LLaMA-3.1:** API-based pipeline for fast inference

5. Results

5.1. Scoring Accuracy

All systems demonstrated strong alignment with human grading (Table 1). The API-based pipeline achieved higher accuracy (R^2 = 0.920, MAE = 0.35) compared to the offline GPT-2 model (R^2 = 0.895, MAE = 0.42). Figure 2 shows the distribution of differences between predicted and actual scores for the GPT-2 model, with most predictions clustering close to the ideal line.

5.2. Inter-Rater Agreement

Cohen's κ statistics (Table 2) showed substantial agreement between model predictions and human scores across all configurations.

Table 1: Core accuracy metrics across models

Model	R^2	MAE	RMSE	Within 0.5	Within 1.0
Offline GPT-2	0.895	0.42	0.54	68.0%	90.0%
GPT-4o + LLaMA-2	**0.920**	**0.35**	**0.48**	**72.0%**	**93.0%**
Groq LLaMA-3.1	0.910	0.37	0.50	70.0%	92.0%

Table 2: Inter-rater agreement (Cohen's κ)

Model	Cohen's κ
Offline GPT-2	0.78
GPT-4o + LLaMA-2	**0.82**
Groq LLaMA-3.1	0.81

5.3. Feedback Quality Assessment

Human evaluation of generated feedback was conducted by three independent raters using a structured rubric assessing specificity, groundedness, and helpfulness on a 5-point Likert scale. The evaluation revealed significant advantages for API-based models (Table 3). The GPT-4o + LLaMA-2 configuration scored highest on specificity (4.3/5), groundedness (4.5/5), and helpfulness (4.4/5). Inter-rater agreement for feedback quality assessment was substantial (Fleiss' $\kappa = 0.75$).

Table 3: Feedback quality ratings (Likert 1-5)

Model	Specificity	Groundedness	Helpfulness
Offline GPT-2	3.2	3.4	3.3
GPT-4o + LLaMA-2	**4.3**	**4.5**	**4.4**
Groq LLaMA-3.1	4.1	4.3	4.2

5.4. Computational Efficiency

The offline model offered superior throughput (2000 scripts/hour) with minimal per-script cost after training (Table 4). API-based models incurred higher latency (6.5-12.0 seconds per script) and operational costs but required no training infrastructure.

Table 4: Runtime and efficiency comparison

Model	Avg. time/script (s)	Throughput (/h)	Cost/script (USD)
Offline GPT-2	**1.8**	**2000**	**0.00**
GPT-4o + LLaMA-2	12.0	300	0.15
Groq LLaMA-3.1	6.5	550	0.10

6. Limitations

We faced several limitations during this study. Since we were required to obtain the students' explicit consent for processing their data, the dataset was relatively small (64 students' exams). The dataset is from a single Data Science course. Although it can be used to demonstrate feasibility, having this small dataset limits us from measuring

the system performance across different fields, subjects, and educational contexts. Furthermore, having all responses graded by a single instructor could introduce grading bias. Consequently, this makes it difficult to determine whether the models are learning standard assessment practices or replicating instructor grading biases from that single evaluator. Utilizing GPT-4o Vision as an OCR component may introduce optical character recognition (OCR) errors, which were not measured in this study. Moreover, the dataset contained exams only in English. Processing exam papers in other languages may significantly affect system performance, from the OCR component through the entire pipeline. The API-based models depend on using external services which may introduce latency and ongoing costs.

7. Discussion

7.1. Strengths of Automated Grading Approach

The systems demonstrated strong scoring accuracy with MAE below 0.42 across all configurations on a dataset of 64 student papers. API-based models achieved Cohen's κ above 0.80, indicating substantial agreement with human raters. The integration of RAG frameworks improved feedback groundedness and reduced hallucinations. Calibration analysis showed minimal systematic bias across score ranges. The scalability of the system was demonstrated through processing 256 pages of exam content efficiently.

7.2. Implications for Deployment

A hybrid strategy is recommended. Utilizing local (offline) models for routine, short questions and using API systems for complex answers that need high quality feedback. Institutions should consider privacy requirements, computer resources, and feedback quality needs when choosing deployment methods. Human-in-the-loop verification is crucial for high-risk assessments. The successful processing of 64 student papers shows that this method works for medium sized courses.

8. Conclusion

This study discusses utilizing transformer-based models for short answer grading automation and generating explanatory feedback, using a dataset of 64 student papers from the Current Topics in Data Science course. Although the local (offline) GPT-2 model offers an efficient solution for institutions with privacy concerns or customization needs, the API-based pipeline provides higher feedback quality, albeit with higher operational costs and reliance on external services.

9. Future Work

Future research should address several important areas. Enriching the dataset by diversifying subjects, institutions, languages, and evaluators would enhance model reliability, improve generalizability, and reduce potential bias. Developing customized OCR models for educational hand-written purposes can also reduce the recognition errors, with quantitative error rate reporting. Implementing bias detection and mitigation mechanisms is essential to ensure fair evaluation. Integrating hybrid workflows, which use different methods dynamically based on complexity, would save resources on a large scale.

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